বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি, কুমিল্লা
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Department of Computer Science and Engineering

Lab Report

Lab Report No	:	**03**
Lab Report Name	:	**A* Search Algorithm (Romania Road Map Problem)**
Course Title	:	Artificial Intelligence Sessional
Course Code	:	CSE 422
Name	:	MD Rakibul Hassan
ID	:	0822220105101024
Level : **4** **Term** : **2** **Section** : **A** **Group** : **G1**		
Date of Submission : **17/05/2026** **Semester** : Spring **Year** : 2026		

Marking Rubric

Problem Understanding & Report Clarity (3)	Implementation (4)	Results, Analysis & Presentation (3)	Total (10)

Key Learnings:

- Learned about the A* search algorithm with hands-on experience.
- Understand the working process of the A* search algorithm, how it combines actual and heuristic values to reach the goal quickly.

Code Implementation & Sample Input/Output:

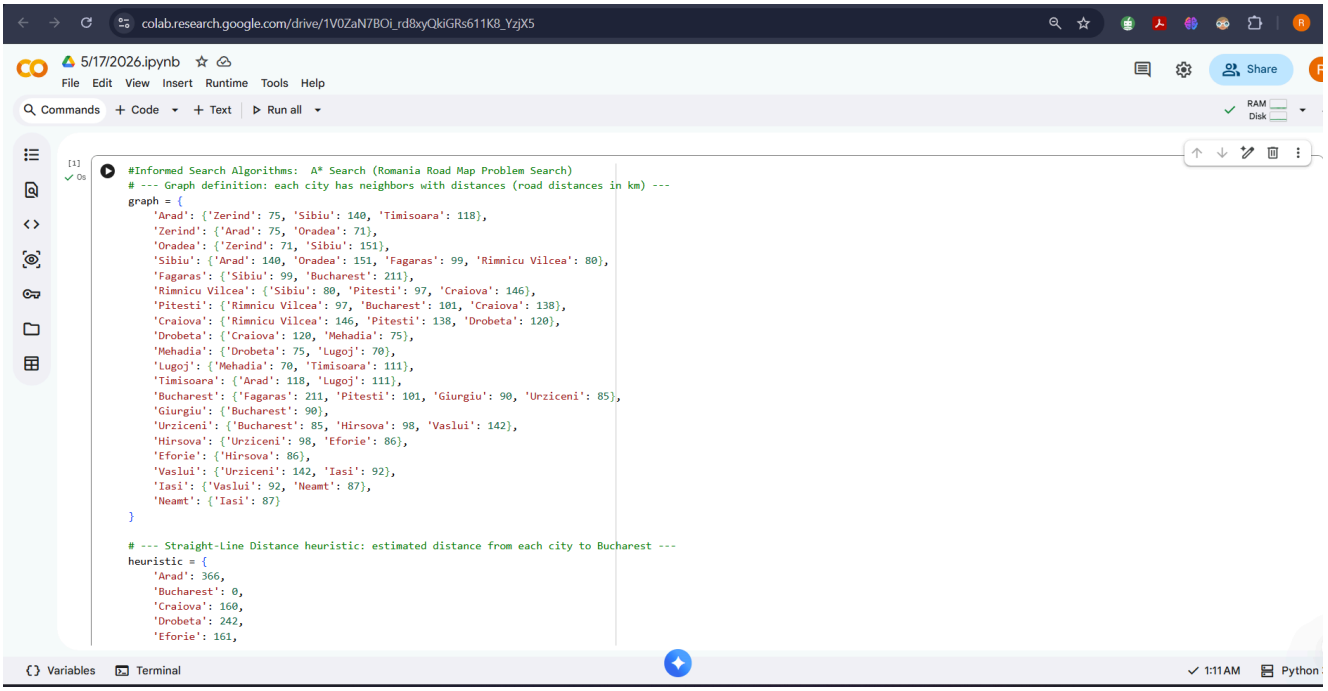

Figure No: 01

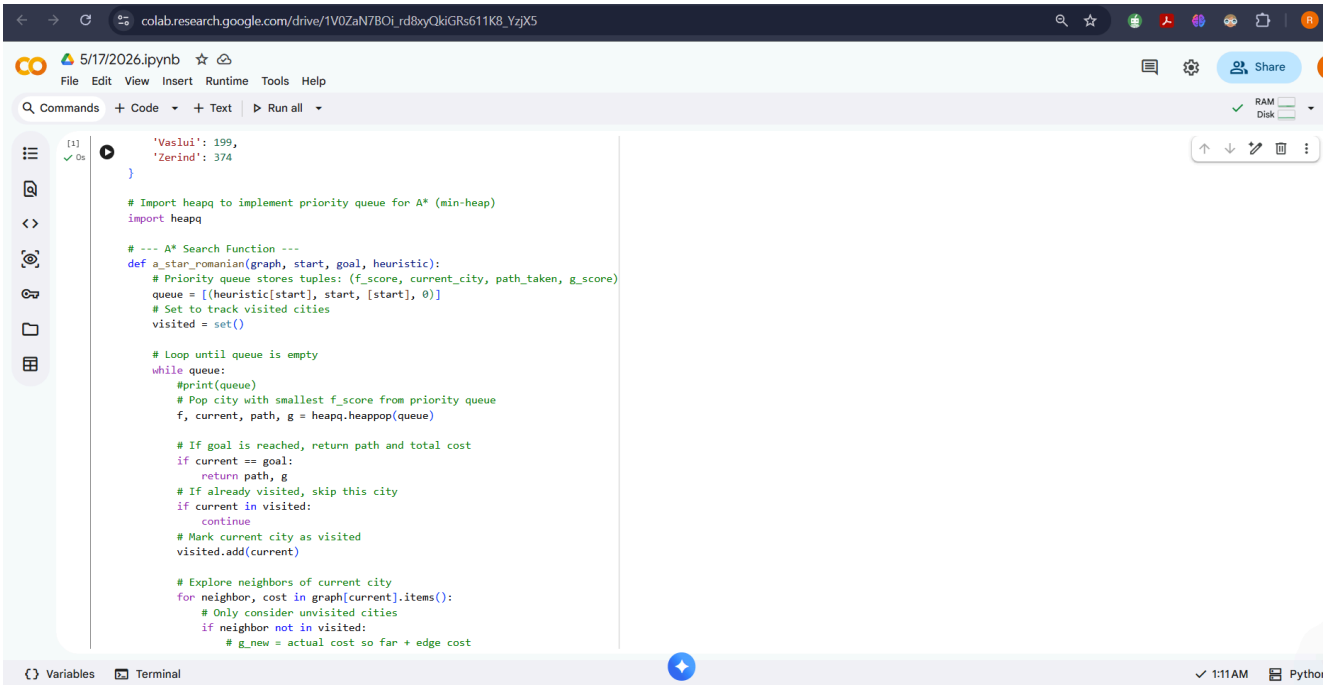

Figure No: 02

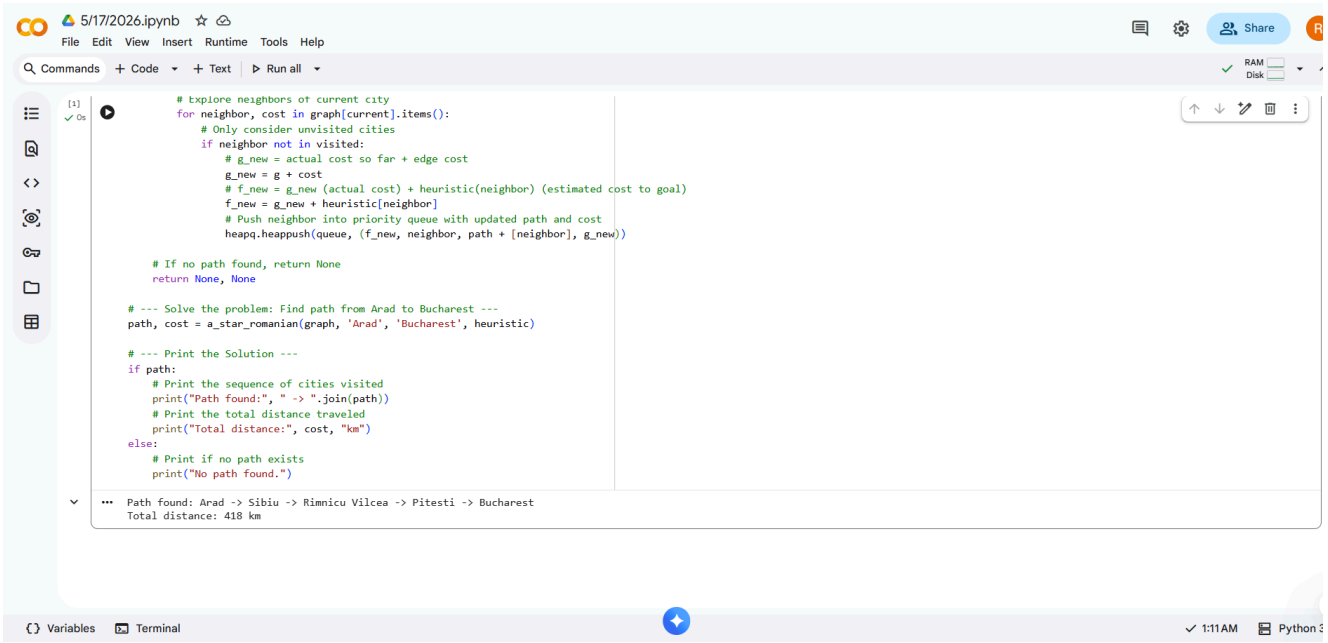

Figure No: 03

Result Analysis

- **The A* search algorithm is one of the popular techniques that are used for path-finding and graph traversal.**
- **A* search is an informed search algorithm that ignores the visited track to avoid the same search repeatedly.**
- **A* search successfully gets the shortest path from Arad to Bucharest, which is optimal.**